

Protocol on the Evaluation of the Knowledge-based Multi-Agent Architecture Design (MAAD) Framework

1 Protocol

1. **Introduction:** Software architecture design is a critical, knowledge-intensive phase that requires balancing competing quality attributes and adapting to evolving requirements. Traditionally, this process is time-consuming, heavily reliant on expert judgment, and often yields limited design alternatives under agile delivery pressures. While Large Language Model (LLM)-based agents have shown promise across various software engineering tasks, their application to automated architecture design remains underexplored, particularly regarding domain-aware reasoning, systematic evaluation, and multi-view consistency. To address these gaps, we proposed MAAD (Multi-Agent Architecture Design), a knowledge-driven framework that orchestrates four specialized agents (*Analyst*, *Modeler*, *Designer*, and *Evaluator*) to collaboratively transform requirements into comprehensive architectural blueprints. MAAD integrates Retrieval-Augmented Generation (RAG) to infuse authoritative standards and design patterns, and employs a hierarchical memory mechanism for iterative refinement. Through this interview, we aim to gather your expert perspectives on the correctness, practical utility, and adoption considerations of MAAD-generated artifacts in real-world development scenarios.

2. **Open Question:** The open question of our interview is based on the Research Question of our study:

- *RQ1: How effective is MAAD in automating the software architecture design?*

Rationale: If the generations of MAAD are both useful and accurate, it could substantially reduce the time and expertise required in practice. RQ1 evaluates the usefulness and correctness of MAAD-generated artifacts via expert feedback, and then uncovers deployment challenges and practitioner concerns.

3. **Participants:** We will conduct in-depth interviews with practitioners who have rich industrial architecture design experience. During each interview, we will:

- Review sample artifacts produced by MAAD and solicit feedback on their correctness, clarity, and practical utility.
- Discuss any limitations or issues observed.
- Elicit suggestions and identify the principal challenges they foresee when integrating LLM-based agents into existing architecture workflows.

4. **Data Analysis:** Interview transcripts will be analyzed using open coding and the Constant Comparison method¹. This qualitative approach will enable us to identify recurring themes, categorize practitioner perspectives, and refine our understanding of both MAAD’s strengths and its deployment challenges.

¹S. Adolph, W. Hall, and P. Kruchten, “Using grounded theory to study the experience of software development,” *Empirical Software Engineering*, vol. 16, no. 4, pp. 487–513, 2011.

2 Interview

Interview Title: Interview on Evaluation of a Knowledge-based Multi-Agent Architecture Design (MAAD) Framework

Problem Statement: Traditional software architecture design remains heavily dependent on human expertise, often leading to cognitive overload, inconsistent decision-making, and scalability bottlenecks. Architects must navigate complex trade-offs among quality attributes (e.g., performance, security, maintainability) while adapting to evolving constraints and agile lifecycles. This process is inherently knowledge-intensive and relies on tacit experience, particularly when entering unfamiliar domains. Consequently, architects often struggle to rapidly explore diverse design alternatives or systematically validate architectural conformance. While existing multi-agent systems (MAS) excel at code generation, they lack architecture-specific workflows, domain knowledge integration, and structured quality evaluation mechanisms. Automating and augmenting this process with knowledge-driven, feedback-aware AI is essential to accelerate architecture formulation while preserving design integrity, traceability, and standard compliance.

Our Approach: We proposed MAAD, a knowledge-driven multi-agent framework that automates the end-to-end architecture design workflow. MAAD orchestrates four role-specific LLM agents: the *Analyst* (requirements decomposition into FRs, NFRs, and ASRs), the *Modeler* (multi-view “4+1” architectural modeling in PlantUML), the *Designer* (production-ready documentation synthesis), and the *Evaluator* (iterative quality validation, cross-view consistency checking, and overall evaluation). To ground generation in established best practices, MAAD employs RAG over a curated knowledge base of authoritative standards (e.g., ISO/IEC/IEEE 42010/42020) and architectural textbooks, complemented by a three-layer hierarchical memory mechanism (working, episodic, and semantic) for cross-iteration learning. Rather than aiming for fully autonomous replacement, MAAD is designed as an intelligent assistant that transforms architecture design into a structured, feedback-driven process. It reduces manual validation overhead and mitigates hallucinations while maintaining human-in-the-loop oversight for contextual judgment, tacit knowledge integration, and final architectural sign-off.

Interview Questions

Open-ended Questions (OQs):

OQ1: How would you assess the correctness and practical value of the MAAD framework’s outputs—specifically the requirements analysis, architectural views, generated documentation, and evaluation report? To what extent do you believe these artifacts could support or accelerate a real-world architecture design?

(Free text)

OQ2: Based on your experience, what are the main challenges in deploying LLM-powered agents for automated architecture design? What strategies or improvements would you recommend to make such a framework more effective and trustworthy in practice?

(Free text)

OQ3: MAAD integrates knowledge from existing system designs, scholarly literature, and expert input. To what extent does the framework effectively leverage these diverse knowledge sources? What enhancements to MAAD’s knowledge-integration process would enable it to better emulate the cognitive processes of human architects?

(Free text)